# Exploring the role of everyday language in comprehending mathematical concepts: A case study of sets and infinity

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*The use of mathematical terms in everyday life affects students' mathematical understanding. In this study, we examined the effect of everyday language on students' understanding of the concepts of set and infinity. The participants were 6th, 9th, and 12th-grade students from Greece. What emerged from the data was that, the everyday meaning of these terms had influenced students' understanding of these mathematical concepts and that some of the misconceptions that lower-grade students had were also found in higher grades students.*

*Keywords: Infinity, set, language.*

## Introduction

Words with specific meanings in mathematics may have different meanings in daily life. Students usually use some mathematical terms in everyday life before they are taught their meaning in mathematics. The concepts of set and infinity are fundamental in Mathematics and crucial in mathematics education. In the Greek mathematical curriculum, the concept of set is introduced, but not given sufficient focus despite its frequent use in the classroom. The concept of infinity, although used, especially in higher grades, is never formally introduced. Recent studies have highlighted the importance of language-responsive mathematics teaching in multilingual classrooms (Cirillo et al., 2010; Prediger, 2022). In this study, we investigate the impact of everyday language usage on the understanding of the terms set and infinity, when the language of instruction is the same as the spoken language, but these terms are being used in everyday Greek language with a different meaning than in formal mathematics language.

## Theoretical framework

A register is a set of meanings that is appropriate to a particular function of language along with the words and structures which express these meanings (Halliday, 1978). Words with specific meanings in the *formal mathematics register* may have different meanings in the *everyday life register*. The *school mathematics register* is a mixture of everyday language and formal mathematics language that differs from both of them. Mathematics teachers have to act as multilinguals that try to bridge the gap between those three registers and consciously pass their linguistic expertise to their students (Gough, 2007). Mathematics teachers cannot assume that the language required to access mathematical learning has been properly developed in other curriculum classrooms, such as language, or in daily life (Adoniou & Qing, 2014).

Different and sometimes conflicting views exist regarding linking the everyday language register to the formal mathematics language register, as well as the methods for facilitating this linking. Prediger (2019; 2022), for instance, views it as a one-way progression from informal everyday expressions to school academic mathematical language and eventually to formal mathematical language. In contrast, Barwell (2012, p.332) argues that this is an "instrumental view of the role of language in learning and

development, in which mathematical discourse is primarily understood as a tool for mathematical thinking", and proposes a more nuanced perspective that regards the relationship between everyday language and mathematical language as a dynamic interplay, where each register enhances the other. That being said, there are certain words that have fundamentally different meanings in each register and cannot always be connected. The everyday usage of such words can create misconceptions and hinder students' comprehension of the corresponding mathematical concepts.

One way to take advantage of the interplay between the everyday language and the mathematical language is by promoting approaches to mathematics teaching that are responsive to students' language use. Despite the widespread accepted need for language-responsive mathematics teaching, only a small number of teachers are adequately prepared for this task (Prediger, 2019). Although language-responsive classrooms have faced criticism for their exclusive emphasis on vocabulary as a stand-alone objective, a clear understanding of the necessary components of language-responsive mathematics teaching has not yet emerged (Prediger, 2019). It is important to include exposure to etymologies, or word origins, because they help build bridges between everyday language and mathematical language (Cirillo et al., 2010; Thompson & Rubenstein, 2000). Vocabulary is very important for comprehension of specific concepts. As we progress through the school grades, the vocabulary of mathematics becomes increasingly idiosyncratic to the learning area and as the concepts become more abstract so does the vocabulary. Words are used precisely in mathematical texts. Sometimes these words are discipline-specific and not encountered in any other learning area, but often they are words that may be used differently in everyday speech (Cirillo et al., 2010). Words that have a unique usage in the formal mathematics register can create tensions between different registers.

In axiomatic set theory, the set is a primitive concept, which means that there is no formal definition, but the definition taught in schools is that of a collection of discrete and well-defined objects. In Greek, the word that is being used in mathematics for the *set* concept is "sýnolo" (a compound word composed of the prefix syn meaning with or together and the word ólo which means all or whole). According to the dictionary of Babiniotis (2012), the mathematical term set originates from the French word *ensemble* while the linguistic term has the same meaning as the English word *group*. It can be used to describe people or objects put together constituting a whole (treated as a unit because of a common property or characteristic that connects them), or a combination of things that fit together (e.g. combination of clothes). That being said, in everyday colloquial Greek language, a set is being used mostly as *total* (for modern terms, it is a semantic loan both from the French word ensemble and also the French word totalité) or as the *outcome* (usually the sum).

In Greek, the word for *infinity* is "apeiro", which means "without end". This word can only be used in daily life metaphorically or as an exaggeration. For example, a student might complain that the teacher assigned them an infinite amount of exercises. According to Babiniotis (2012), the word infinity was used as a philosophical term in early cosmology, identifying matter as the origin of the world and as not subject to spatiotemporal dimensions. It is that which has no end, that lasts forever, that is not bounded by spatial and temporal limits, that is so large or so many that it cannot be measured or counted. In the school mathematics register the word infinity is used in various contexts and in various ways (as a process, as an object etc.) without ever being defined, thus, students are

even more likely to be influenced by the everyday, metaphorical use of the term. According to Monaghan (2001) students usually use the word infinity as an adjective (potential infinity) and when they use it as a noun, it is hard to discern which meaning they assign to it (process or object).

Prediger (2022) identified several research goals that researchers can choose to pursue, with regards to typical problems in mathematics education. One of those goals is "describing" and "explaining" a specific problem with language in mathematics. In this study, we focus on "describing" and "explaining" the effects of the everyday Greek language on students' understanding of the concepts of set and infinity. It is our supposition that acknowledging the potential limitations imposed by everyday language usage is a prerequisite when attempting to implement strategies for language-responsive mathematics teaching.

## Methodology

Our research question for this study was: how does the use of everyday language affect students' understanding of the concepts of set and infinity? The present study is a collaborative effort between researchers who conducted two distinct investigations into students' understanding of the concepts of infinity and set during school education. Greek school education has three stages. Primary school (grades 1-6), middle school (grades 7-9), and high school (grades 10-12). In those studies, the researchers administered a questionnaire to 398 students, including 132 $6^{th}$ graders, 153 $9^{th}$ graders, and 113 $12^{th}$ graders, to investigate their conceptions of infinity. Similarly, the researchers administered another questionnaire to 397 students, including 116 $6^{th}$ graders, 127 $9^{th}$ graders, and 154 $12^{th}$ graders, to investigate their conceptions of set (Vlachos, 2022). We selected students from those age groups (i.e., $6^{th}$ graders, $9^{th}$ graders, and $12^{th}$ graders) to represent the final stage of their respective educational level (i.e., primary school, junior high school, and high school). Upon analysing the quantitative data from the questionnaires administered to these students, we observed that most students had developed their own conceptions of infinity and set. To investigate the origins of these conceptions further, we conducted semi-structured, semi-clinical interviews (Elkind, 1964) with a sample of six students from each grade level who had completed the infinity questionnaire, and five students from each grade level who had completed the set questionnaire. All participants in this study were either native Greek speakers or bilinguals that were born in Greece and for whom Greek was one of their main languages. All materials used in the study were presented in Greek. The researchers combined some of their findings from these qualitative studies and viewed them through a language lens, in an attempt to "describe and explain" (Prediger, 2022) the effects of everyday language on students' understanding of the concepts of set and infinity.

## Results

In this segment, we present the conceptions of students from $6^{th}$, $9^{th}$ and $12^{th}$ grades, regarding the concepts of set and infinity, as they emerged from the interviews. Our aim is to observe how the everyday usage of the words 'set' and 'infinity' in Greece influenced the formation of the corresponding mathematical concepts in these students' minds. Initially, we describe the conceptions regarding the concept of set, held by three students at different grade levels: Victor, a $6^{th}$ grader, Penny, a $9^{th}$ grader, and Charles, a $12^{th}$ grader. In the interview with Victor, we gave him a hypothetical discussion between a teacher and her students in which the teacher asked whether one's

pencils, erasers, books and snacks can constitute a set. One hypothetical student answered "yes, I have 3 pencils, 1 eraser, 4 books and 1 snack therefore I have 9 objects set (meaning in total), and another disagreed saying that "the teacher didn't asked how many objects you have, the correct answer is they don't constitute a set because snack is food and everything else are objects we use in the classroom." Victor agreed with the first student, so we asked him what he thinks a set is. He replied that "it is how many objects is there, for example on a table, the number of objects on the table." Later, in the same interview, the following dialog emerged:

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| Researcher: | Suppose you have the set {1,2,3,4} and the set {0,2,3,5}, are those equal? |
| Victor: | Yes they are equal. |
| Researcher: | Why? |
| Victor: | Because you've added 1 and 4 which makes 5 and you've put 0 instead of 1. |
| Researcher: | So, what you are saying is that, if you add the numbers up, you get the same outcome? |
| Victor: | Yes. |
| Researcher: | So maybe a set is not the number of objects but rather the sum? |
| Victor: | Yes. |
| Researcher: | In the first scenario you had pencils and snacks. How are you going to add them? |
| Victor: | Yes, it was the number of objects in that case, but it is the sum now because we have numbers. |

In his first response, Victor used the Greek word for set (which is sýnolo) as the *total amount* of objects and in his second response he used it as the *sum*. Both those meanings are valid in the everyday Greek language but are completely different from the meaning of the word "set" in mathematics. Victor's understanding of the set concept seems to have been influenced by these uses of the word "set" in the Greek language, so he understands the concept of set in mathematics both as a sum and as a number of objects, depending on the context.

In the interview with Penny, we gave her a hypothetical discussion between a teacher and a student in which the teacher asked whether the digit 1, the digit 2 and the letters a and b constitute a set, and the student answered that they can constitute a set because there are two numbers and two letters so these are four objects in total (the Greek word used for "in total" was the same as the word used in Greek for "set"). We asked Penny if she agrees with this student's answer and the next dialogue followed:

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|---|---|
| Penny: | I feel like, the student is confused. In my mind a set is something different. I don't perceive the set as how many objects there are. I feel like the set is something different. |
| Researcher: | Something different, like what? |
| Penny: | That it must be similar things... people e.g. the Greeks constitute a set because they are all Greeks. |

In the above dialogue, Penny seems to understand the concept of set as a collection of *similar* objects, a condition that is not necessary for the objects of a set. But this usage of the term set is common in everyday life (the phrase "social set" is commonly used colloquially in Greek daily life interchangeably with the word *society*). Later, in the same interview, we asked her whether the sets {1,2,3,4} and {5,6,7,8} are equal and she answered "no, because if we add them up they don't have the same outcome." In this instance, it is possible that Penny adopted another usage of the Greek word for set, that of the sum. Much like Victor, two conflicting conceptions regarding the notion of

set seem to coexist in Penny's mind, both of which are derived from the everyday usage of the word in Greece.

In the next episode, we gave Charles, a 12th grader a hypothetical scenario between a teacher and her students. The teacher asked her students "what is the set of the letters of the word tennis." One of the hypothetical students answered that "the set of the letter of the word tennis is 6", a second hypothetical student stated that "the teacher didn't ask how many letters are there, the correct answer is {t,e,n,i,s}" and a third student replied that "in order to create the word tennis you need all the letters, so the correct answer is {t,e,n,n,i,s}." Charles answered that he agrees with the first hypothetical student and the next dialogue followed:

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|---|---|
| Researcher: | Why wouldn't you say that the word tennis is composed of the letters t, e, n, i and s so I put those into… |
| Charles: | Because, in effect, she said *the set* (meaning the number of objects) of the letters of the word, she didn't say *which* letters are included in the word. |
| Researcher: | So, in what sense do you use the word set here? |
| Charles: | Set… the set of the word that the teacher gave in order to find out, I don't know… how many letters it has. |

In the excerpt above, Charles emphasized the word "set", contrasting that with the word "which". In his view, a set is the number of objects and not the collection of the objects themselves. Reading Charles argument in English might seem misplaced but, in everyday Greek, it makes perfect sense.

Following, we describe the conceptions about the infinity of Dennis, Gloria and Larry, who were 6th, 9th and 12th graders respectively.

In the excerpt below is our discussion with Dennis who stated that people have an infinite amount of hair on their heads:

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| Researcher: | You said that people have an infinite amount of hair on their heads? |
| Dennis: | I remember having a discussion with my classmates and some of them definitely said "yes because we can't count them." |
| Researcher: | OK, so take my head for example. Do you consider those hair to be infinite or just a very large number that we don't have the ability to count? |
| Dennis: | I think both of them, in a way... because we can't count them and it is a very, very large number. |
| Researcher: | If there were a special device that could count the hair on our heads and it gave you an enormous number, would you say they are infinite? |
| Dennis: | No. |
| Researcher: | OK and then you said that there are not infinite browned haired women in Greece. |
| Dennis: | No, there aren't, it's the same as with the hair, if we had a special device that could count the brown-haired women then... |
| Researcher: | So you answered "no" because it is easier to count women? |
| Dennis: | Yes, but, also because women are not like hair that can fall and regrow in a second. |

As we mentioned above, infinity has two main uses in the everyday Greek language. The first is something that has no end, limits or boundaries and the second is something extremely large. Dennis seems to have been influenced by both uses of the term and understands infinity, on one hand as an endless perpetual process ("hair that can fall and regrow"), and on the other hand as something that is either too large to be counted or that we lack the ability to count it ("because we can't count them and it is a very, very large number").

In the interview with Gloria, a 9th grader, we asked her to give an example for the word “infinity”:

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| Gloria: | Infinite cells. |
| Researcher: | Are they actually infinite, or in the sense that there are many cells? |
| Gloria: | They are infinite. |
| Researcher: | In what sense? |
| Gloria: | Because we can’t count them, and they are being reborn. |
| Researcher: | So you said “infinite” because it’s something that we are unable to count and that it doesn’t stop, as a process? If there was a magical way to count the cells would you still believe that they are infinite? |
| Gloria: | Yes |
| Researcher: | Why? You don’t trust the counting or because they are being reborn? |
| Gloria: | Because we can’t count them precisely, one will always be missing. |

In the above excerpt, Gloria constructed a conception of infinity based on what she learned in a biology class. For her, infinity is something that cannot be counted and that has no end. She was taught that a human has an enormous number of cells that are constantly dying and being reborn. The vastness of cells in our bodies and their perpetual cycle of death and rebirth lead her to assume that there is an infinite number of cells in the human body.

The last episode is an excerpt from the interview with Larry, who is a 12th grader. During the interview he argued that there is an “infinite amount” of grains of sand on earth, not just some huge number. Then the next dialogue followed:

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| Larry: | I believe that grains of sand are infinite because we will never be able to count them, there are far too many. |
| Researcher: | So you use infinity in cases when we don’t have the ability to count something? |
| Larry: | Yes. |
| Researcher: | I see, so for you what is infinity? Is it an inability to count something? |
| Larry: | Something that is so big, so much, that we cannot count it by any means. Let’s say, we can’t count all the grains of sand on the earth. |
| Researcher: | If technology advances and they finally provided us with a number, a huge number, would you still claim that they are infinite? |
| Larry: | If we had a number? |
| Researcher: | Yes, if we could count them with a special device. |
| Larry: | I wouldn’t consider them infinite, no. |
| Researcher: | It’s like saying that there are two kinds of infinity, numbers, which they never end, and things that are too many and we haven’t developed the necessary technology to count them. What is more prevalent to you? |
| Larry: | Both, because technological development might provide further insight on infinity. |

In this case, the student employs the term "infinity" to refer to a quantity that is immensely large and cannot be counted. He acknowledges having two different conceptions of infinity, that of the endless and that of a quantity that we cannot measure. When posed with a hypothetical scenario where a device could count the sand grains, he suggested that the quantity would no longer be infinite. This may indicate that he understands that it is not truly infinite and that he is using the word metaphorically.

## Discussion

Certain mathematical terms are used in everyday language not only in their mathematical sense but also in other ways. Sometimes the non-mathematical usage may have no direct or even indirect connection to the mathematical concept, or it may even refer to a different mathematical concept

altogether. In the Greek language, two examples of such terms are “set” and “infinity”. The Greek word for set (sýnolo) can mean the number of specific objects, the sum of specific numbers, as well as a collection of similar objects. The Greek word for infinity (apeiro) means “without end”, and in everyday language, more often than not, this word is used to express something that is too much or too big and it is difficult or impossible to count it.

There are different and sometimes conflicting views regarding linking the everyday and formal mathematical language (Prediger, 2019; 2022; Barwell, 2012). In the case of “set” and “infinity” in the Greek language, some of the meanings do not relate to the corresponding mathematical concepts that students may use these words for. Therefore, our study followed Prediger’s (2019; 2022) approach and focused only on the one-way progression, which is the influence of everyday language (informal) on students’ understanding of the concepts of set and infinity (formal). The results of our study indicate that the meanings of these words in the Greek language have an impact on students’ comprehension of the corresponding mathematical concepts. Moreover, as illustrated in the case of Gloria, the use of a word in another field (in our case, biology) can also influence students’ understanding of mathematics. Consequently, a student may have developed multiple and conflicting interpretations of these mathematical concepts, with one or the other prevailing depending on the context. Based on these findings, we agree with Cirillo et al. (2010) that, exposing students to etymologies and word origins can help bridge the gap between the everyday language register, the school mathematics register and the mathematical language registers.

Our findings suggest that everyday language exerts a strong influence on students’ understanding of the concepts of “set” and “infinity”, which is evident in students in the last grade of primary school (grade 6), in the last grade of middle school (grade 9) and the last grade of high school (grade 12). It would be interesting for a future study to investigate why students’ conceptions of these concepts often remain rigid over time, despite continued exposure to the terms in different contexts and to a more nuanced approach of their use in higher educational levels. Moreover, exploring the extent to which students’ mathematical understanding of these concepts affects the use of corresponding terms in everyday language could be a fruitful avenue for future investigation.

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